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Semiconductor Production Equipment | Japan

SEMICON Taiwan Day2 - Heterogeneous Integration Seminar Takeaways

On Day 2 of SEMICON Taiwan 2026, we attended the Heterogeneous Integration Seminar. Here we summarize mainly the keynote presentations.

The international SPE expo SEMICON Taiwan 2026 kicked off Aug 31 (pre-show forums Aug 31-Sep 1, exhibits Sep 2-4). On the second day of the expo (Sep 1), we attended the first day of the Heterogeneous Integration Global Summit. Keynote sessions included speakers from TSMC and AMD, while SK Hynix spoke on HBM. There was lots of talk about topics like 2.5D packaging (CoWoS), 3DIC (hybrid bonding), and PLP.

According to AMD, the number of reasoning tokens has surged 158x over the past two years as agentic AI has become more prevalent, and computational loads required for training are increasing 5x a year. AMD says optimization at the system level is needed to improve data center energy efficiency as electricity consumption is widely known to have become a critical bottleneck amid rapid growth in compute demand. AMD targets a 20x improvement in rack-scale energy efficiency in 2024-30. Achieving that will require a combination of measures such as 3D stacking, PLP, and optical interconnects (in addition to steps at the rack level like cooling methods).

**TSMC is covered by Charlie Chan, AMD by Joseph Moore, SK hynix by Shawn Kim.*

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SEMICONDUCTOR PRODUCTION EQUIPMENT

Japan

Industry View

Attractive

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TSMC said it put 5.5x-reticle CoWoS into mass production in 2026, and is steadily attaining yields of 98%+ (5.5x reticle CoWoS-L is currently used in products like the AMD MI455 (5.7x). TSMC apparently plans to target 9x reticle in 2027, and 14x further down the line.

The power used per AI chip package has grown to 6x the level in 2021 as packages have become bigger, and power delivery has become a challenge due to a rise in the energy density within packages. In its speech, TSMC also covered integrated voltage regulators (IVRs) and other components embedded within the interposer to address this problem. A non-negligible reduction in voltage occurs due to phenomenon like parasitic resistance and parasitic inductance in the wiring when high voltages are transmitted within packaging, and housing voltage regulators conventionally placed on PCB within the interposer shortens the wiring length to the extent possible and significantly alleviates the problem noted above.

As package sizes increase, the industry will need to shift to panel-level packaging (PLP) and adopt even larger panels. According to AMD, a circular 300mm wafer currently yields 16 modules, but larger package sizes could reduce this to just three modules in the future. Even the 310mm × 310mm square panels currently under development for CoPoS yield no more than six modules, while a 510mm square panel could yield 16. AMD explained that a shift to larger panels will become essential to maintain the number of modules produced per substrate.

Larger panels should also create more business opportunities for maskless exposure systems that support fine pitches, including Ushio's DLT system and Nikon's DSP-100. In its latest April–June results, Ushio announced that some customers had selected its system for mass-production use and expects full-scale volume production to begin from F3/29 onward.

For D2D hybrid bonding, TSMC currently offers SoIC solutions supporting 9 μm pitches for N7 and 6 μm pitches for N5/N3/N2. It plans to support volume production at a 4.5 μm pitch for A14 in 2029. When asked about the challenges associated with hybrid bonding, TSMC identified the CMP process as the most critical. The process must planarize the dissimilar materials of the Cu pads and surrounding interlayer dielectric film with extremely high precision, making it technically challenging. Sony Semiconductor Solutions (Sony is covered by Kazuo Yoshikawa), has also highlighted the importance of CMP in hybrid bonding. Precise control of Cu recess in the Cu-CMP process, which finishes the Cu electrode surface several nanometers below the surrounding interlayer dielectric film, is critical.

In CMP for advanced packaging, Ebara says it has gradually increased its share and has secured at least 50% of the market. Given its experience in CMP for CIS hybrid bonding, we believe Ebara can also expand its business opportunities in 3D ICs.

SK hynix explained why hybrid bonding is needed in HBM. It wrapped up reasons for introducing hybrid bonding in HBM, including (1) enabling maintenance of JEDEC standards' package height limit (720/775 μm) by moving to a bumpless design, (2) significantly expanding bandwidth by dramatically increasing the number of IOs and wiring

density with fine-pitch wiring under 1 μm , and (3) substantially reducing thermal resistance (by 30% vs MR-MUF). However, SK hynix said it plans to make the most of highly cost-effective MR-MUF where it has a track record and it is primarily using Advanced MR-MUF for 16 Hi HBM4/4E, only mentioning with regard to hybrid bonding that it is pursuing development on that front at the same time. Also, the company said that it is true that some customers are concerned about heat generation in response to a question about moves to reduce the number of layers per stack.

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Overweight/Buy	1555	42%	450	48%	29%	767	44%
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Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/01/2026)
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DISCO (6146.T)	O (07/07/2023)	¥56,590
Ebara (6361.T)	O (12/04/2024)	¥4,570
JEOL (6951.T)	E (07/08/2025)	¥7,110
KOKUSAI ELECTRIC (6525.T)	O (12/19/2025)	¥8,559
Lasertec (6920.T)	U (07/28/2025)	¥33,080
Nikon (7731.T)	U (08/08/2023)	¥1,956
SCREEN Holdings (7735.T)	O (11/13/2025)	¥13,480
Tekscend Photomask (429A.T)	E (11/18/2025)	¥3,725
Tokyo Electron (8035.T)	O (12/19/2025)	¥54,980
Ulvac (6728.T)	E (01/17/2024)	¥7,385
Ushio (6925.T)	O (01/05/2026)	¥3,857

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